

Bola Hosny Refaat

DevOps Engineer

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SUMMARY

Computer Science student with hands-on experience in AWS cloud infrastructure and DevOps fundamentals gained through internships and intensive training programs, including NTI and the Digital Egypt Pioneers Initiative (DEPI). Skilled in Linux, Docker, Git, and cloud architecture, with practical exposure to designing scalable and secure solutions through real labs and projects. Motivated to grow as a Cloud/DevOps Engineer and contribute to real-world cloud systems.

EDUCATION

- Faculty of Computers and Information at Future Academy Cairo, Egypt
- Major: Computer Science Grade: Very Good
- 2022 -2026

EXPERIENCE & INTERNSHIP

- **[Digital Egypt Pioneers Initiative - DEPI - DevOps Engineer trainee] (Nov 2025-Present)**
 - Currently training on DevOps trainee learning Linux, Bash scripting, Git, and GitHub.
 - Learning containerization using Docker and environment setup for applications.
 - Participating in structured labs and coursework as part of the DEPI DevOps track. **[Online training]**
- **[Sprints x Microsoft Summer Camp - DevOps Foundations Internship] (Aug 2025-Sep 2025)**
 - Completed an intensive 40+ hour, 3–4-week training in DevOps fundamentals, including software development processes, containerization, and AWS Cloud & Microsoft Learn essentials.
 - Applied concepts through practical capstone projects, enhancing technical skills and aligning with industry best practices. **[Online training]**
- **[Cloud Architect] summer training at (National Telecommunication Institute) (Aug 2025-Sep 2025)**
 - Completed 120-hour, 4-week training focusing on virtualization concepts and public cloud access via AWS.
 - Solved 10+ hands-on labs and knowledge checks, applying core cloud concepts in practical scenarios.
 - Gained in-depth knowledge of AWS cost solutions, resource management, and designing application architectures to meet technical business objectives. **[Online training]**.
- **[Infrastructure Engineer trainee] at (Suez Canal Authority) (Jul 2025-Aug 2025).**
 - Participated in hands-on training on IT infrastructure systems, including network setup, server maintenance, and hardware configuration.
 - Gained practical experience managing enterprise-level infrastructure and supported tasks related to LAN/WAN design, system monitoring, and troubleshooting **On-Site [Ismailia, Egypt]**.
- **[Aws cloud] summer training at (National Telecommunication Institute) (Aug 2024-Sep 2024)**
 - Solved 10+ hands-on labs and knowledge checks, applying core cloud concepts in practical scenarios.
 - Completed a 4-week, 120-hour program focusing on foundational AWS services, cloud architecture, and deployment. **[Online training]**
- **[Flight Simulator training] summer training at (Egypt Air) (Jul 2024-Aug 2024).**
 - Gained insights into the engineering aspects of aviation simulation, including system operations, hardware integration, and real-time data analysis. **On-Site [Cairo, Egypt]**.

COMPETITIONS AND ACHIEVEMENTS

- Competitive programming – Solved over 150+ algorithmic problems on **[codeforces]**.
- Participated in the ECPC (**Egyptian Collegiate Programming Contest competition**) (2024-2025)
- achievement as one of the Top Graduates of the "Summer Training Program" Organized by the (ITIDA) In collaboration with (NTI). (2024-2025)

SKILLS

- **Programming Languages:** C++, Python, SQL.
- **Tools and concepts:** Problem Solving, Data Structures, Algorithms, OOP, AWS, Git, GitHub, Docker, Kubernetes, Bash.
- **Operating Systems:** Windows, Windows server, Linux: [Ubuntu, Red Hat].
- **Languages:** Arabic (Native Language) -- English (Very Good at Reading and Writing.)

COURSES

- VMware: [Cloud & Virtualization Concepts - Network Virtualization Concepts -Software-Defined Storage].
- Amazon Web Services (AWS): [AWS Academy Cloud Foundations - AWS Academy Cloud Architecting].
- KodeKloud: [Learning Linux Basics Course & Labs - Docker Training Course for the Absolute Beginner].
- Cisco Networking Academy: [Linux Essentials Certification].
- The Linux Foundation: [Introduction to Kubernetes].
- Huawei ICT Academy: [HCIA-Datacom V1.0 Course].

PROJECT

- **AWS Academy Cloud Architecting – Capstone Project (National Telecommunication Institute)**
 - Engineered and deployed a **highly available PHP application** on AWS with **Application Load Balancer** and **Auto Scaling Groups**, ensuring scalability and fault tolerance.
 - Designed and secured **Amazon RDS MySQL** with private subnets, **Secrets Manager** integration, and strict access controls.
 - Demonstrated advanced expertise in **AWS cloud architecture, security hardening, and enterprise-grade solution design.**
- **Emotion Recognition System (CNN-based) (future Academy):**
 - Developed a real-time facial emotion detection model using CNNs (TensorFlow, Keras, OpenCV).
 - Trained on 35k+ images (FER2013), achieving 95% training and 82.8% validation accuracy.
 - Optimized over 5 iterations with transfer learning and data augmentation, reducing training time by 20% and improving generalization by 30%. Integrated with webcam feed to detect 7 emotions in real-time with <100ms inference latency.
- **Simple Flight Reservation Desktop Application (Self Project)**

A Python-based desktop app built with Tkinter and SQLite to manage flight bookings.

 - Engineered a **fully functional GUI application** using Tkinter for smooth navigation and user experience.
 - Integrated **SQLite database** to handle secure data storage and implement CRUD operations.
 - Developed core features: **flight search, reservation, and cancellation** workflows.
 - Applied **Python OOP principles** for scalable and maintainable code structure.
 - Strengthened expertise in **desktop application development, database integration, and UI/UX design.**
- **Bash-Based Database Management System**
 - Built a **Bash-based CLI DBMS** supporting full CRUD operations, data validation, and primary key constraints.
 - Enabled system-wide execution through **Linux environment configuration** with structured terminal output.